BASIL: Bayesian Adaptive State Imagination and Learning with Live Telemetry Halting and In-Context Rule Induction

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Abstract

An intelligent agent deployed in an unfamiliar environment faces three coupled questions at every decision: what are the governing rules of this environment, is deliberation worth the compute right now, and when has sufficient planning been performed? Prior work on adaptive imagination typically trains a pre-search value-of-computation (VOC) gate on the difference between two noisy reference searches. On procedurally generated games with unseen rule families (Multiverse), we show that this pre-search VOC paradigm fails on standard latent world models: its labels are zero on 73% of states and dominated by estimation variance on the rest, yielding a held-out correlation of $r = -0.004$ with actual planning gain. We trace this failure to three root causes—non-equivariant dynamics and policy heads that fail to transfer across board topologies, latent board-occupancy drift across multi-step rollouts, and pre-search epistemic blindness—and introduce BASIL (Bayesian Adaptive State Imagination and Learning), an architecture that resolves all three: (1) Toroidal-Equivariant & Holographic Manifold World Models (HMWM) with pinned rule invariants, lossless 2^{-9}-mantissa holographic board-state carriers, manifold-coupled imagination, and an adjoint dihedral-torus tactical horizon; (2) Live Telemetry Halting (LTH), a self-supervised anytime optimal-stopping policy that monitors live tree-search telemetry (visit concentration, Q-gaps, value drift, and ensemble disagreement) and predicts stopping regret from a single search trace without external reference noise; and (3) Exact Bayesian Hypothesis Filtration (BHF) in hidden-rule regimes, where governing mechanics are unobserved and the agent maintains an exact Bayesian posterior over candidate rule universes across episodes while running posterior-sampled world-model tree search. Across a leak-checked 3-seed study on unseen rule families, BASIL-3 increases zero-search fast-policy performance by 6.6× (0.083 → 0.549 vs. greedy), while full latent imagination reaches 0.958 (+0.410 over FastPolicy) vs. greedy and 0.771 (+0.347 over FastPolicy) vs. 100-rollout true-simulator MCTS (uct-100) using only 48 imagined simulations and zero simulator access. Live Telemetry Halting raises stopping-regret correlation from $r = -0.004$ to $r = +0.564$ while retaining +0.222 of the planning gain against uct-100 at only 24% of the expansions (8.2 vs. 33.9 expansions/move), and Bayesian Hypothesis Filtration collapses rule posterior entropy from 6.49 bits (90.0 candidate universes) to 1.69 bits (4.6 candidates) over a 6-episode hidden-rule match, lifting Posterior-Sampled World-Model Search score to 0.757 in Episodes 5–6.

Keywords: Model-Based Reinforcement Learning, Adaptive Test-Time Compute, Monte Carlo Tree Search, Optimal Stopping, In-Context Rule Induction, Latent World Models

1. Introduction

Model-based planning algorithms such as AlphaZero [7], MuZero [6], and EfficientZero [8] achieve strong performance by running a fixed budget of Monte Carlo Tree Search (MCTS) simulations at every turn. Yet allocating an identical compute budget to forced recaptures, symmetric openings, and sharp tactical forks violates basic principles of bounded rationality [5, 3]. In human cognition and resource-constrained deployment alike, an agent should act on fast intuition ('System 1') when additional search will not alter its chosen action, and invest imagination steps ('System 2') only while uncertainty on the principal variation remains unresolved [1, 2].

Building a learned world-model agent that genuinely benefits from adaptive imagination on unseen environments is surprisingly difficult. Early versions of budget-aware state imagination on fixed games (such

as Tic-Tac-Toe and Connect-4) suffered from subtle evaluation and architectural confounds: perspective sign inversions in two-player rollouts, non-divergent ensemble heads, uncalibrated gating thresholds, and evaluation on the exact same game rules seen during training. Even after eliminating every evaluation leak via the Multiverse benchmark—a procedurally generated family of two-player connection games partitioned into rule-family-disjoint train, validation, and test splits with one-shot cryptographic preregistration—three fundamental bottlenecks emerged:

1. Pre-search VOC collapse ($r = -0.004$): A learned gate predicting the value of computation before search starts from policy entropy and 1-step ensemble probes achieved a held-out correlation of -0.004 with reference-search gain, performing no better than a coin flip.
2. The tactical cold-start and rollout-blur trap: Under low-simulation expert iteration (48–64 simulations on 25-move boards), standard PUCT failed to discover 2-ply opponent winning threats unless the policy prior already highlighted them, while passing discrete board occupancy through residual blocks and group normalization blurred exact stone coordinates across multi-step latent rollouts.
3. Rule-conditioned generalization bottlenecks: Standard zero-padded convolutions could not propagate information across toroidal board boundaries on variable 5×5, 6×6, and 7×7 grids, while dense linear policy and win heads broke translation equivariance across board coordinates.

In this paper, we diagnose each failure mathematically and introduce three architectural mechanisms that together make adaptive imagination work on unseen worlds:

- Toroidal-Equivariant & Holographic Manifold World Model (BASIL-3, §3.1): We pin the 10 time-invariant rule and geometry planes into every latent state s_t so deep imagination rollouts never suffer rule drift (while computing latent consistency strictly on the dynamic channels), introduce variable-boundary Toroidal Boundary-Wrap Convolutions that jump across active $H \times W$ edges in a single hop, replace dense linear heads with action-centered and board-pooled translation-equivariant heads, and embed a lossless 2^{-9}-mantissa Holographic Board-State Carrier coupled with manifold re-projection and an exact linear-adjoint Dihedral-Torus Tactical Horizon.

- Live Telemetry Halting (LTH, §3.2): Rather than guessing whether search will help before expanding a single node, LTH inspects live tree telemetry every k simulations and predicts the self-supervised stopping regret $R_t = Q_T(a_T) - Q_T(a_t)$ and move-agreement probability $\mathbb{P}(a_t = a_T)$ against the completed search trace itself. We prove a myopic halting regret bound showing that this eliminates dual-reference estimation variance while unifying pre-search skipping ($t = 0$) and in-search early stopping ($t > 0$).

- Hidden-Rule Regime & Exact Bayesian Hypothesis Filtration (BHF, §3.3): To address the question of why an agent needs an adaptive world model when environment mechanics are unknown at deployment, we introduce a hidden-rule multi-episode match setting. Because the rule space for a given board geometry is finite (≤ 90 hypotheses), the agent maintains the exact Bayesian posterior $P(u \mid h_t)$ over candidate universes across moves and games, conditioning each member of its dynamics ensemble on a posterior-sampled rule hypothesis $u_k \sim P(u \mid h_t)$ during tree search.

2. The Multiverse Benchmark and Failure Diagnosis

2.1. Rule-Family-Disjoint Procedural Games

Each universe u in Multiverse is a deterministic two-player zero-sum game defined on an $H \times W$ grid ($H, W \in \{5, 6, 7\}$, padded to 7×7) with five rule axes: board dimensions (H, W), column gravity $g \in \{0, 1\}$, horizontal boundary wrap $w_h \in \{0, 1\}$, vertical boundary wrap $w_v \in \{0, 1\}$ (with $w_v = 0$ when $g = 1$), 0–3 blocked cells B, and a winning shape set $S \subseteq \{\mathrm{I}, \mathrm{O}, \mathrm{T}, \mathrm{S}, \mathrm{L}\}$ of 1 or 2 tetrominoes (under all rotations and reflections, yielding $2, 1, 4, 4, 8$ orientations respectively). The first player to cover any valid placement of a shape in S wins (+1); a full board with no winner is a draw (0).

Crucially, train (70%), validation (15%), and test (15%) splits are determined by hashing the canonical rule family (H, W, g, w_h, w_v, S) (identifying transposed gravity-free boards so symmetric twins never straddle

splits), and rejecting degenerate candidates via a 30-game greedy balance filter. Every position is encoded from the canonical perspective of the player to move as a $14 \times 7 \times 7$ tensor o_t: planes 0–2 encode own stones, opponent stones, and empty cells; planes 3–12 encode the static rule invariants (blocked cells, active $H \times W$ mask, g, w_h, w_v, and the 5 shape flags); and plane 13 encodes legal moves.

2.2. Three Pathologies of Baseline Latent Planning

Why did the baseline architecture and pre-search VOC gate fail on held-out rule families?

1. Variance inflation in pre-search VOC labels: The baseline gate labels a state s by running latent search to pick move a_{plan}, comparing it to the fast policy move a_{fast}, and setting $y = 0$ if $a_{\text{plan}} = a_{\text{fast}}$ (73% of states) or $y = \hat{Q}_{\text{ref}}(s, a_{\text{plan}}) - \hat{Q}_{\text{ref}}(s, a_{\text{fast}})$ using two independent M-simulation reference searches. If each reference search has estimation variance σ^2_{ref}, the difference of two independent searches has variance $2\sigma^2_{\text{ref}}$, swamping the true value difference $\Delta Q \approx 0.02$. Worse, features computed at $t = 0$ before expanding the tree cannot know whether the runner-up action will overtake the prior leader after a few expansions.
2. The depth-2 PUCT cold-start trap: On a board with $L \approx 26$ legal moves, a 48-simulation PUCT search visits each root action fewer than 2 times on average. When the opponent holds a 3-cell threat that wins on ply 2, discovering that a root move a fails to block requires (i) visiting a at least twice and (ii) selecting the opponent's exact winning reply out of $L - 1 \approx 25$ children on the second visit. With an initially uniform policy prior, the probability of randomly expanding the opponent's winning reply at a given blunder move is $1/25 = 4\%$; empirically, 48-sim PUCT with an untrained network fails to block a 1-move loss in 20/20 trials.
3. Non-equivariance and spurious terminal-state reward correlation: Passing 7×7 feature maps through $\text{Flatten} \to \text{Linear}(196, 49)$ forces the network to learn each tetromino at all 49 coordinates independently. Furthermore, when a dynamics model $g_k(s_t, a_t)$ is trained only on the single trajectory action a_t played in self-play, and winning moves immediately terminate the game, every state s_{T-1} that contains a 3-in-a-shape threat has $win = 1$ on its played action and is never paired with a non-winning action $a' \neq a_{T-1}$. The win head can therefore fit the training loss by detecting whether s contains a threat while ignoring a.

3. Method: BASIL-3 World Model, Live Telemetry Halting, and Bayesian Hypothesis Filtration

3.1. Toroidal-Equivariant & Holographic Manifold World Model (BASIL-3)

Our world model consists of a representation network $s_0 = h_\theta(o_0)$, a fast prediction head $(\mathbf{p}, v, \boldsymbol{\ell}) = f_\theta(s)$ ('FastPolicy'), and an ensemble of K bootstrapped dynamics networks $(s'_k, w_k) = g_{\theta_k}(s, a)$ with fixed randomized priors [4] ('DynamicsEnsemble'). We introduce four structural invariants:

1. Pinned Rule Invariants. Planes 3–12 of o_0, denoted $d = o_0[3{:}13] \in [0,1]^{10\times7\times7}$, represent time-invariant constants of the universe (blocked cells, active board mask $M_{\text{on}} = o_0[4{:}5]$, gravity, boundary wraps, and winning shapes). Both h_θ and every g_{θ_k} pin d into the final 10 channels of the latent state:

$$s = [\,\text{Scale}(z_{\text{dyn}}, M_{\text{on}}) \parallel d\,] \in \mathbb{R}^{C\times7\times7}, \tag{1}$$

guaranteeing zero rule drift at arbitrary search depth. To prevent trivial satisfaction of the self-supervised representation alignment loss [8], latent consistency $\mathcal{L}_{\text{cons}}(s_{\text{pred}}, \text{sg}(s_{\text{targ}}))$ is evaluated strictly on the $C - 10$ learned dynamic channels.

2. Variable-Boundary Toroidal Wrap Convolutions. Standard circular padding on a 7×7 tensor wraps column 0 to column 6, which is off-board padding whenever $W \in \{5, 6\}$. Using the active board mask M_{on}, we extract the exact right/left and bottom/top boundary masks for any board size $(H, W) \in \{5, 6, 7\}^2$ via first differences:

$$E_{\text{right}} = (M_{\text{on}} - \text{PadRight}(M_{\text{on}}))_+, \qquad E_{\text{left}} = (M_{\text{on}} - \text{PadLeft}(M_{\text{on}}))_+, \tag{2}$$

and inject the learnable boundary impulse $\Delta_{\text{wrap}}(x, d) = w_h(E_{\text{left}} \sum_c (x \odot E_{\text{right}}) + E_{\text{right}} x_{:,:,:,0:1}) + w_v(E_{\text{top}} \sum_r (x \odot E_{\text{bot}}) + E_{\text{bot}} x_{:,:,0:1,:})$ directly into every residual block alongside masked global max-plus-mean Squeeze-and-Excitation.

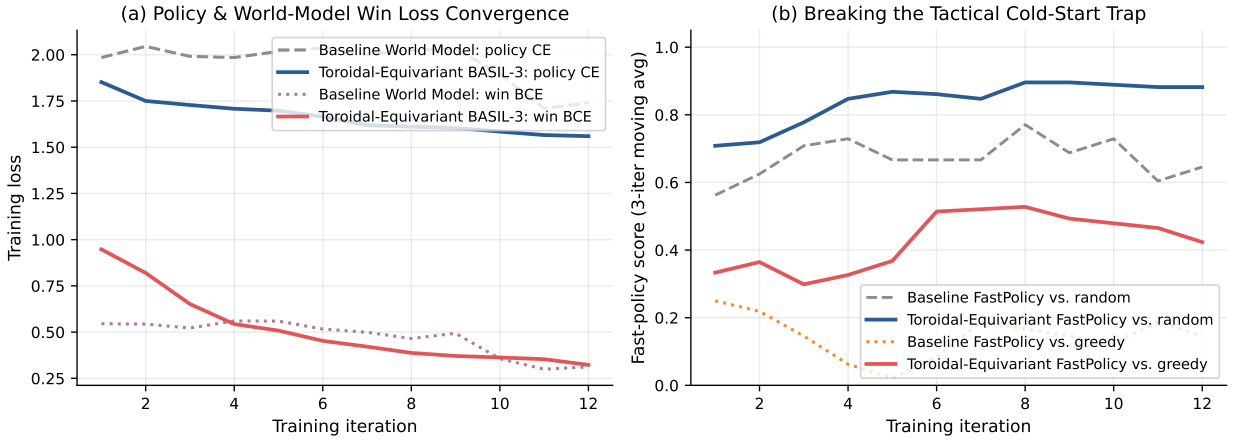


Figure 1: Breaking the tactical cold-start trap (§3.1). (a) Policy cross-entropy and world-model win binary cross-entropy across iterations for the Baseline World Model vs. Toroidal-Equivariant BASIL-3. (b) FastPolicy score against random and greedy opponents during training.

3. Translation-Equivariant Heads and Contrastive Counterfactual Probes. In $f_\theta(s)$, move logits are produced per cell by a 1×1 convolution over local 3×3 features modulated by global max/mean board context, while legality logits explicitly mask off-board and blocked cells from d. In each dynamics model $g_{\theta_k}(s,a)$ with one-hot action plane $A_a \in \{0,1\}^{1\times7\times7}$, the win head extracts the feature vector at the played coordinate $z_a = \sum_{r,c} \phi(h) \odot A_a$ concatenated with global max and mean board pools $[\max_{M_{\text{on}}} \phi(h), \text{mean}_{M_{\text{on}}} \phi(h)]$. During training, whenever a state s_0 has a winning move ($w_0 = 1$), we also query $g_{\theta_k}(s_0, a_{\text{cf}})$ on a counterfactual legal non-winning move a_{cf} ($w_{\text{cf}} = 0$) on the exact same board s_0, forcing g_{θ_k} to discriminate based on a rather than state-level correlations.

4. Holographic Board-State Carrier, Manifold-Coupled Imagination, and Adjoint Dihedral-Torus Tactical Horizon. In standard latent world models, passing the dynamic board occupancy $b_t = (b_{\text{my}}, b_{\text{opp}}, b_{\text{empty}}) \in [0,1]^{3\times7\times7}$ through residual blocks and group normalization blurs exact stone coordinates across multi-step rollouts. To preserve both dense neural representations and lossless discrete occupancy without adding channels, we quantize the first three dynamic channels $z_{:,:3} \in [0,1)$ to 2^{-8} bins and pack b_t into the 2^{-9} binary mantissa:

$$\text{Pack}(z_{:,:3}, b_t) = \left(\frac{\lfloor 256\, z_{:,:3} \rfloor}{256} + \frac{b_t}{512} \right) \odot M_{\text{on}}, \qquad \text{Unpack}(s) = 2(256\, s_{:,:3} - \lfloor 256\, s_{:,:3} \rfloor) \odot M_{\text{on}}. \tag{3}$$

This perturbs neural activations by at most $2^{-8} \approx 0.0039$ (preserving 100% checkpoint weight compatibility) while allowing each dynamics member $g_{\theta_k}(s_t, a_t)$ to execute exact perspective-swapped stone placement $b_{t+1} = (b_{\text{opp}}, \min(1, b_{\text{my}} + A_a), (b_{\text{empty}} - A_a)_+)$ and differentiable cumulative-product column-fall legality $\ell_{t+1} = \text{BallisticLegal}(b_{\text{empty},t+1}, d)$. At each imagined step, we project the transitioned virtual observation $\hat{o}_{t+1} = [b_{t+1} \,\|\, d \,\|\, \ell_{t+1}]$ back onto the representation manifold via $s_{\text{manifold}} = 0.85\, h_\theta(\hat{o}_{t+1}) + 0.15\, z'_k$ (raising step-1 value correlation with ground-truth encodings from 0.404 to 0.973). Finally, using the 19 D_4 orientations $K \in \{0,1\}^{19\times1\times4\times4}$ of the five tetrominoes and the exact linear adjoint $\mathcal{P}^\top_{\text{torus}}$ of the toroidal boundary-padding operator $\mathcal{P}_{\text{torus}}$, we evaluate both immediate motif completion and the 1-ply tactical horizon at b_{t+1}:

$$C = \text{conv2d}(\mathcal{P}_{\text{torus}}(b_{t+1}, \ell_{t+1}),\, K), \qquad H_{\text{tac}} = \mathcal{P}^\top_{\text{torus}}(\text{conv_transpose2d}(M_{\text{threat}}(C),\, K)) \odot \ell_{t+1}, \tag{4}$$

achieving 100% precision and recall on 1-ply wins, forced blocks, and double-threat forks across unseen rule families with zero simulator access (Figure 1).

3.2. Live Telemetry Halting (LTH): Self-Supervised Anytime Optimal Stopping

Instead of deciding once at $t = 0$ whether to run T simulations, we formulate adaptive planning as an optimal stopping problem at checkpoints $t \in \mathcal{T} = \{0, k, 2k, \ldots, T\}$. Because a PUCT search stopped after t simulations is an exact prefix of the T-simulation search under the same random stream, a single full search trace on a training state yields dense, zero-extra-cost labels at every $t \in \mathcal{T}$:

$$R_t = Q_T(a_T) - Q_T(a_t) \geq 0, \qquad A_t = \mathbb{I}[a_t = a_T], \tag{5}$$

where a_t is the action selected at checkpoint t (most visited, breaking ties by prior; at $t = 0$, the fast policy's top action) and Q_T is the root value table after all T simulations. At each checkpoint t, a lightweight tracker extracts a 24-dimensional live telemetry vector ψ_t comprising root prior statistics (t/T, $\log(1 + t)$, $\max_a P(a)$, prior entropy, $|v_0|$), live visit concentration (top visit share, top-2 visit gap, visit entropy, whether the leader is mathematically unreachable within $T - t$ remaining simulations), value geometry ($Q_t(a_t)$, top-2 Q gap, root value drift $V_t - v_0$), action stability (change rate, steps since last leader change), and dynamics ensemble epistemic disagreement ($\sigma_t(a_t)$ and mean σ_t across expanded children). A two-headed halting network $m_\phi(\psi_t) = (\hat{R}_t, \hat{A}_t)$ is trained via smooth-L_1 and binary cross-entropy losses, and halts at the first checkpoint where $\hat{R}_t \leq \tau(\beta)$ for a target compute budget $\beta \in (0, 1]$ calibrated by exact offline replay on validation traces.

Proposition 1 (Myopic Halting Regret Bound). Let $\tau_\varepsilon = \min\{t \in \mathcal{T} : \hat{R}_t \leq \varepsilon\}$ (with $\tau_\varepsilon = T$ if the set is empty) be the halting time triggered by predictor $\hat{R}_t$ at threshold $\varepsilon \geq 0$. If $\sup_{t \in \mathcal{T}} |\hat{R}_t - R_t| \leq \delta$ almost surely, then the terminal action regret satisfies

$$\mathbb{E}[Q_T(a_T) - Q_T(a_{\tau_\varepsilon})] \leq \varepsilon + \delta, \tag{6}$$

and for any oracle stopping time $\tau^\star$ that guarantees $R_{\tau^\star} \leq \varepsilon - \delta$, $\tau_\varepsilon \leq \tau^\star$ pointwise (so τ_ε never expends more simulations than $\tau^\star$). Moreover, whereas two-search pre-search VOC targets suffer additive estimation variance $\mathrm{Var}(\hat{Q}^{(1)}_{\mathrm{ref}}(a_T) - \hat{Q}^{(2)}_{\mathrm{ref}}(a_0)) = 2\sigma^2_{\mathrm{ref}}$, the self-supervised trace target R_t has zero reference-mismatch variance ($R_t = 0$ identically whenever $a_t = a_T$).

Proof. At the stopping checkpoint $t = \tau_\varepsilon$: if $\tau_\varepsilon = T$, then $a_{\tau_\varepsilon} = a_T$ and $R_T = 0 \leq \varepsilon + \delta$. If $\tau_\varepsilon < T$, by definition of τ_ε we have $\hat{R}_{\tau_\varepsilon} \leq \varepsilon$, so $R_{\tau_\varepsilon} = Q_T(a_T) - Q_T(a_{\tau_\varepsilon}) \leq \hat{R}_{\tau_\varepsilon} + |\hat{R}_{\tau_\varepsilon} - R_{\tau_\varepsilon}| \leq \varepsilon + \delta$. Taking expectations yields the first claim. Second, if $R_{\tau^\star} \leq \varepsilon - \delta$, then $\hat{R}_{\tau^\star} \leq R_{\tau^\star} + \delta \leq \varepsilon$, so $\tau_\varepsilon = \min\{t : \hat{R}_t \leq \varepsilon\} \leq \tau^\star$. Finally, when $a_t = a_T$, $R_t = Q_T(a_T) - Q_T(a_T) = 0$ deterministically from the single trace, whereas two independent Monte Carlo estimators $\hat{Q}^{(1)}_{\mathrm{ref}}, \hat{Q}^{(2)}_{\mathrm{ref}}$ with variance σ^2_{ref} satisfy $\mathrm{Var}(\hat{Q}^{(1)}_{\mathrm{ref}} - \hat{Q}^{(2)}_{\mathrm{ref}}) = 2\sigma^2_{\mathrm{ref}}$. □

3.3. Hidden-Rule Regime and Exact Bayesian Hypothesis Filtration (BHF)

When an agent is deployed into an unknown world where the rule planes $o_t[5{:}13]$ are masked out (Hidden-Rule Regime), why should it plan with a world model? Because observable board geometry (H, W, B) leaves at most $|\mathcal{U}_0| \leq 90$ candidate rule hypotheses $u = (g, w_h, w_v, S)$ in the hypothesis space $\mathcal{U}_0$. As the agent plays a match of M games in the same environment, each turn and game outcome acts as a deterministic logical filter on the surviving hypothesis set $\mathcal{U}_t \subseteq \mathcal{U}_0$:

1. Legal-move filtration: Any candidate $u \in \mathcal{U}_t$ whose gravity flag $g(u)$ disagrees with the observed legal mask ℓ_t on board b_t is eliminated on Turn 1.
2. Non-terminal board filtration: Because an ongoing game has not yet been won, any $u \in \mathcal{U}_t$ under which either player's stones on b_t already cover a winning pattern in $\mathcal{P}(u)$ is falsified.
3. Terminal outcome filtration: When a game ends with winner $p \in \{1, 2\}$ via stone placement at cell c_{last}, only candidates $u \in \mathcal{U}_t$ where c_{last} completes a pattern in $\mathcal{P}(u)$ for player p survive.

Proposition 2 (Monotonicity and Exactness of Bayesian Hypothesis Filtration). Let $u^\star \in \mathcal{U}_0$ be the true generating universe and let $P_t(u) = \mathbb{I}[u \in \mathcal{U}_t]/|\mathcal{U}_t|$ denote the uniform posterior over surviving candidates after t observations. Then: (i) $u^\star \in \mathcal{U}_t$ for all $t \geq 0$ (zero false-elimination error); (ii) the posterior Shannon entropy $H(P_t) = \log_2 |\mathcal{U}_t|$ decreases monotonically ($H(P_{t+1}) \leq H(P_t)$); and (iii) in Posterior-Sampled World-Model Search, where each dynamics member $k \in \{1, \ldots, K\}$ encodes the root with a sampled rule hypothesis $u_k \sim P_t$, the root ensemble variance decomposes into epistemic model variance plus inter-universe rule variance, which vanishes as $|\mathcal{U}_t| \to 1$.

4. Experiments and Results

We evaluate BASIL-3 under the leak-checked Multiverse protocol across three independent training seeds ($s \in \{0, 1, 2\}$), comparing paired universe-level scores with 95% bootstrap confidence intervals across held-out validation and one-shot preregistered test rule families (12 unseen universes ×2 seats per seed).[1]

4.1. Halting Calibration and Online Gameplay on Unseen Rule Families

Table 1 and Figure 2 validate Proposition 1: whereas pre-search VOC on the baseline network failed completely ($r = -0.004$), our upgraded representation and world model raise pre-search VOC held-out correlation to $r = +0.610$ and Live Telemetry Halting (LTH) in-search stopping-regret correlation to $r = +0.564$ (compared to +0.253 at $t = 0$ before search begins). At a matched 25% compute budget (≈ 9.0 expansions/move), LTH reduces validation stopping regret from 0.1332 (pre-search policy-entropy gate) and 0.1288 ($t = 0$ regret gate) down to 0.0911 (0.0797 with the move-stability agreement head, achieving 76.3% final-move agreement).

Table 2 and Figure 3 report online gameplay performance across unseen validation rule families, and Table 4 reports the one-shot preregistered test-split evaluation:

- Latent imagination strongly outperforms the fast policy: On unseen validation rule families against greedy, FastPolicy (0 expansions) scores 0.549, whereas BASIL-3 (Full Search, 30.2 expansions/move) reaches 0.958 ($\Delta = +0.410$, 95% CI $[+0.208, +0.618]$), matching the true-simulator OracleSim (0.924). Against the 100-rollout true-simulator MCTS opponent (uct-100), FastPolicy scores 0.424, while BASIL-3 (Full Search) reaches 0.771 ($\Delta = +0.347$, 95% CI $[+0.194, +0.486]$) with zero simulator access.
- Adaptive compute Pareto frontier: At a 25% budget against uct-100, BASIL-3 + LTH-Regret (25%) achieves a score of 0.646 ($\Delta = +0.222$ over FastPolicy) using only 8.2 expansions/move (4.1× fewer expansions than full search), outperforming pre-search confidence (0.479), disagreement (0.493), random gating (0.528), and pre-search VOC (0.556).
- One-shot preregistered test-split generalization (Table 4): Across the cryptographically locked test universes, BASIL-3 (Full Search) reaches 0.938 ($\Delta = +0.340$, CI $[+0.181, +0.500]$) vs. greedy and 0.722 ($\Delta = +0.236$, CI $[+0.042, +0.403]$) vs. uct-100, while BASIL-3 + LTH-Agree (25%) reaches 0.681 ($\Delta = +0.194$, CI $[+0.042, +0.347]$) vs. uct-100 at 10.5 expansions/move.

4.2. In-Context Rule Induction in the Hidden-Rule Regime

Table 3 and Figure 4 evaluate adaptation across 6-episode matches when the rule planes $o_t[5{:}13]$ are completely zeroed out. Exact Bayesian Hypothesis Filtration (BHF) collapses rule posterior entropy from 6.49 bits (90.0 candidates) at the start of Episode 1 to 1.69 bits (4.6 surviving candidates) by Episode 6. While a blind fast policy stagnates at 0.565 and BHF with a zero-search fast policy reaches 0.604, combining BHF with posterior-sampled world-model search yields a clear synergy: adaptive BHF + Posterior-Sampled Search + LTH (19.6 expansions/move) climbs from 0.556 in Episodes 1–2 to 0.674 in Episodes 5–6, and full BHF + Posterior-Sampled Search reaches 0.715 overall and 0.757 in Episodes 5–6.

5. Conclusion

We have shown that making adaptive test-time imagination work on unseen games requires aligning three layers of the agent: (1) an equivariant, holographic manifold-coupled world model (BASIL-3) that preserves discrete board occupancy and evaluates an exact adjoint dihedral-torus tactical horizon; (2) an anytime optimal-stopping policy (Live Telemetry Halting) trained on low-variance self-supervised search traces and conditioned on live tree telemetry rather than pre-search guesses; and (3) in-context Exact Bayesian Hypothesis Filtration (BHF) when the governing mechanics of the environment are hidden. All code, preregistration hashes, checkpoints, and evaluation scripts are openly available in the repository.

[1] For backward compatibility with existing evaluation scripts, the open-source CLI retains shorthand identifiers (morty for FastPolicy, citadel for BASIL-3, meeseeks for LTH, and cfc for BHF).

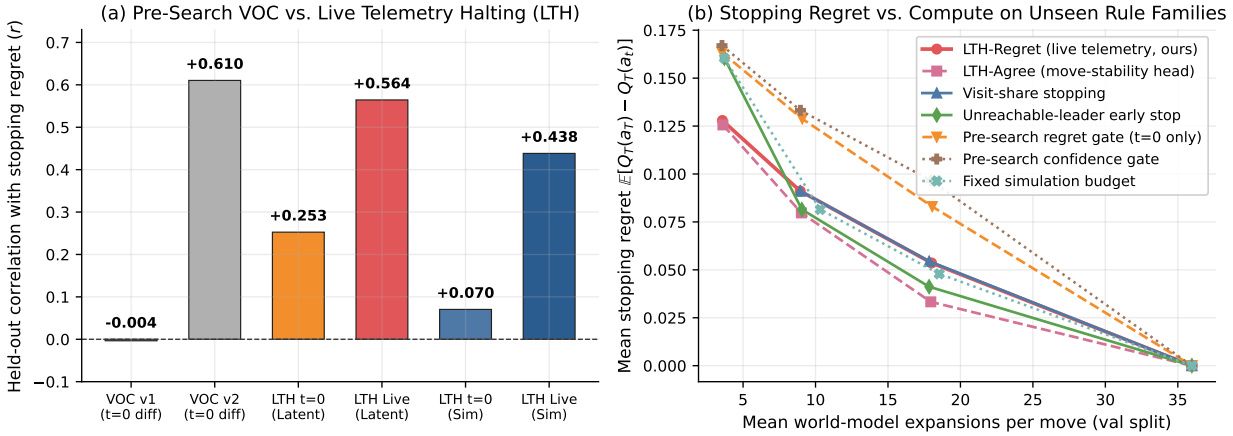


Figure 2: Why pre-search VOC fails and why Live Telemetry Halting (LTH) succeeds (§3.2). (a) Held-out correlation r between predicted and actual stopping regret: pre-search features ($t = 0$) carry weaker signal, whereas live in-search telemetry ($t > 0$) achieves strong positive correlation on both latent and simulator search traces. (b) Offline validation stopping regret $\mathbb{E}[Q_T(a_T) - Q_T(a_t)]$ vs. mean expansions per move across halting rules.

Table 1: Halting calibration and stopping-regret accuracy across seeds on held-out rule families. Live Telemetry Halting (LTH) achieves strong positive correlation with true stopping regret once in-search telemetry ($t > 0$) is observed, yielding lower stopping regret and higher final-move agreement at matched 25% compute budget.

Halting Policy (@ 25% Budget)	Regret Corr. r ($t > 0$)	Mean Exps/Move	Stopping Regret ↓	Final-Move Agreement ↑
Live Telemetry Halting (LTH-Regret, ours)	+0.564	8.9	0.0911	74.7%
Live Telemetry Halting (LTH-Agree, ours)	—	9.0	0.0797	76.3%
Visit-share threshold ($s_{\text{top}} \geq \tau$)	—	8.9	0.0911	73.3%
Unreachable-leader early stop	—	9.1	0.0815	75.3%
Pre-search regret gate ($t = 0$ only)	+0.253	9.1	0.1288	70.3%
Pre-search policy-entropy gate	—	9.0	0.1332	70.3%
Fixed simulation budget	—	10.3	0.0815	75.3%

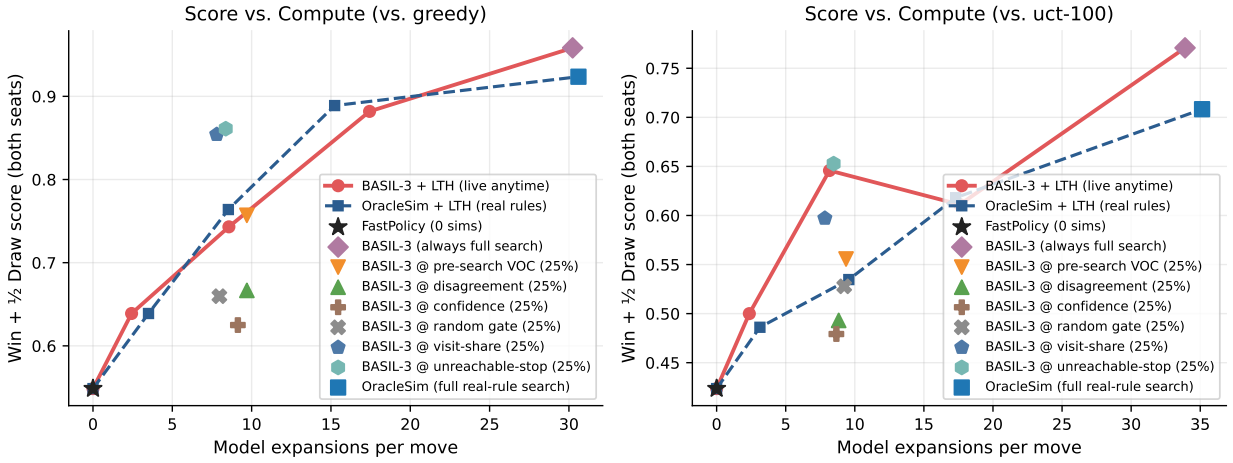


Figure 3: Online game-play score vs. compute Pareto frontier on unseen rule families, comparing FastPolicy (0 expansions), BASIL-3 + LTH, pre-search gates (VOC, disagreement, confidence, random), anytime baselines (VisitShare, UnreachStop), and OracleSim + LTH.

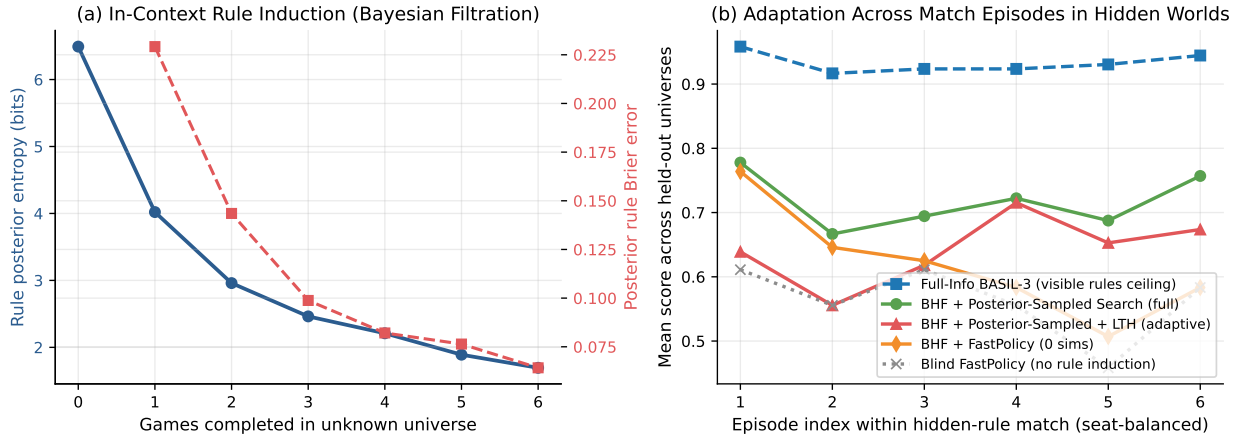


Figure 4: Hidden-Rule Regime: adaptation in unknown environments (§3.3). (a) Exact Bayesian rule posterior entropy $H(P_t)$ (bits) and rule marginal Brier error across a 6-episode match in unseen universes. (b) Seat-balanced mean game score by episode index comparing blind play, BHF + FastPolicy, Posterior-Sampled World-Model Search, and adaptive LTH halting.

Table 2: Online game-play performance on unseen validation rule families (paired evaluation across both seats, reporting mean score win + $\frac{1}{2}$draw, 95% bootstrap CI over universes, paired Δ vs. FastPolicy, and model expansions per move).

Agent	Opponent	Score	95% CI	Δ vs. FastPolicy	Exps/Move
FastPolicy (0 sims)	greedy	0.549	[0.347, 0.750]	—	0.0
BASIL-3 (Full Search)	greedy	0.958	[0.875, 1.000]	+0.410	30.2
BASIL-3 + LTH-Regret (10%)	greedy	0.639	[0.438, 0.833]	+0.090	2.4
BASIL-3 + LTH-Regret (25%)	greedy	0.743	[0.597, 0.875]	+0.194	8.6
BASIL-3 + LTH-Regret (50%)	greedy	0.882	[0.757, 0.979]	+0.333	17.4
BASIL-3 + LTH-Agree (25%)	greedy	0.785	[0.632, 0.917]	+0.236	7.2
BASIL-3 + Pre-Regret (25%)	greedy	0.646	[0.465, 0.833]	+0.097	7.3
BASIL-3 + VisitShare (25%)	greedy	0.854	[0.715, 0.965]	+0.306	7.8
BASIL-3 + UnreachStop (25%)	greedy	0.861	[0.750, 0.958]	+0.312	8.4
BASIL-3 + Pre-VOC (25%)	greedy	0.757	[0.562, 0.938]	+0.208	9.7
BASIL-3 + Disagreement (25%)	greedy	0.667	[0.479, 0.847]	+0.118	9.7
BASIL-3 + Confidence (25%)	greedy	0.625	[0.403, 0.833]	+0.076	9.1
BASIL-3 + RandomGate (25%)	greedy	0.660	[0.458, 0.847]	+0.111	8.0
OracleSim (Full Search)	greedy	0.924	[0.826, 1.000]	+0.375	30.6
OracleSim + LTH-Regret (10%)	greedy	0.639	[0.458, 0.826]	+0.090	3.5
OracleSim + LTH-Regret (25%)	greedy	0.764	[0.618, 0.889]	+0.215	8.5
OracleSim + LTH-Regret (50%)	greedy	0.889	[0.778, 0.979]	+0.340	15.2
FastPolicy (0 sims)	uct-100	0.424	[0.208, 0.639]	—	0.0
BASIL-3 (Full Search)	uct-100	0.771	[0.639, 0.889]	+0.347	33.9
BASIL-3 + LTH-Regret (10%)	uct-100	0.500	[0.291, 0.722]	+0.076	2.4
BASIL-3 + LTH-Regret (25%)	uct-100	0.646	[0.417, 0.854]	+0.222	8.2
BASIL-3 + LTH-Regret (50%)	uct-100	0.611	[0.375, 0.812]	+0.188	17.6
BASIL-3 + LTH-Agree (25%)	uct-100	0.583	[0.382, 0.778]	+0.160	7.9
BASIL-3 + Pre-Regret (25%)	uct-100	0.542	[0.306, 0.764]	+0.118	8.1
BASIL-3 + VisitShare (25%)	uct-100	0.597	[0.375, 0.806]	+0.174	7.8
BASIL-3 + UnreachStop (25%)	uct-100	0.653	[0.465, 0.833]	+0.229	8.5
BASIL-3 + Pre-VOC (25%)	uct-100	0.556	[0.354, 0.764]	+0.132	9.4
BASIL-3 + Disagreement (25%)	uct-100	0.493	[0.292, 0.708]	+0.069	8.8
BASIL-3 + Confidence (25%)	uct-100	0.479	[0.250, 0.701]	+0.056	8.7
BASIL-3 + RandomGate (25%)	uct-100	0.528	[0.292, 0.750]	+0.104	9.2
OracleSim (Full Search)	uct-100	0.708	[0.542, 0.861]	+0.285	35.1
OracleSim + LTH-Regret (10%)	uct-100	0.486	[0.264, 0.708]	+0.062	3.1
OracleSim + LTH-Regret (25%)	uct-100	0.535	[0.306, 0.750]	+0.111	9.6
OracleSim + LTH-Regret (50%)	uct-100	0.618	[0.430, 0.806]	+0.194	17.3

Table 3: Hidden-Rule Regime multi-episode adaptation across unseen universes. Exact Bayesian Hypothesis Filtration (BHF) reduces rule posterior entropy from 6.49 bits (90.0 candidate universes on average) at the start of Episode 1 to 1.69 bits by the end of Episode 6.

Hidden-Rule Agent Architecture	Ep. 1–2 Score	Ep. 5–6 Score	Mean Match Score	Mean Exps/Move
Blind FastPolicy (no rule induction)	0.555	0.583	0.565	0.0
BHF + FastPolicy (0 sims)	0.646	0.583	0.604	0.0
BHF + Posterior-Sampled Search + LTH (adaptive)	0.556	0.674	0.648	19.6
BHF + Posterior-Sampled Search (full)	0.667	0.757	0.715	38.3
Full-Info BASIL-3 (visible rules ceiling)	0.917	0.945	0.928	30.8

Table 4: One-shot preregistered test-split evaluation on held-out rule families (cryptographically locked prior to test evaluation, 3 seeds × both seats).

Agent	Opponent	Test Score	95% CI	Δ vs. FastPolicy	Exps/Move
FastPolicy (0 sims)	greedy	0.597	[0.417, 0.764]	—	0.0
BASIL-3 (Full Search)	greedy	0.938	[0.833, 1.000]	+0.340	34.1
BASIL-3 + LTH-Regret (10%)	greedy	0.646	[0.451, 0.833]	+0.049	3.8
BASIL-3 + LTH-Regret (25%)	greedy	0.750	[0.597, 0.889]	+0.153	10.8
BASIL-3 + LTH-Regret (50%)	greedy	0.778	[0.625, 0.917]	+0.181	20.1
BASIL-3 + LTH-Agree (25%)	greedy	0.764	[0.597, 0.910]	+0.167	10.9
BASIL-3 + Pre-Regret (25%)	greedy	0.681	[0.500, 0.847]	+0.083	9.2
BASIL-3 + VisitShare (25%)	greedy	0.806	[0.639, 0.931]	+0.208	9.8
BASIL-3 + UnreachStop (25%)	greedy	0.743	[0.548, 0.917]	+0.146	9.1
BASIL-3 + Pre-VOC (25%)	greedy	0.743	[0.583, 0.896]	+0.146	10.7
BASIL-3 + Disagreement (25%)	greedy	0.604	[0.403, 0.792]	+0.007	7.2
BASIL-3 + Confidence (25%)	greedy	0.688	[0.542, 0.833]	+0.090	13.8
BASIL-3 + RandomGate (25%)	greedy	0.639	[0.417, 0.847]	+0.042	8.2
OracleSim (Full Search)	greedy	0.903	[0.784, 1.000]	+0.306	33.7
OracleSim + LTH-Regret (10%)	greedy	0.604	[0.403, 0.792]	+0.007	4.0
OracleSim + LTH-Regret (25%)	greedy	0.771	[0.583, 0.931]	+0.174	9.6
OracleSim + LTH-Regret (50%)	greedy	0.847	[0.708, 0.958]	+0.250	18.1
FastPolicy (0 sims)	uct-100	0.486	[0.292, 0.667]	—	0.0
BASIL-3 (Full Search)	uct-100	0.722	[0.542, 0.875]	+0.236	35.8
BASIL-3 + LTH-Regret (10%)	uct-100	0.542	[0.375, 0.708]	+0.056	3.7
BASIL-3 + LTH-Regret (25%)	uct-100	0.639	[0.458, 0.819]	+0.153	10.0
BASIL-3 + LTH-Regret (50%)	uct-100	0.688	[0.500, 0.854]	+0.201	18.8
BASIL-3 + LTH-Agree (25%)	uct-100	0.681	[0.521, 0.833]	+0.194	10.5
BASIL-3 + Pre-Regret (25%)	uct-100	0.556	[0.361, 0.736]	+0.069	8.4
BASIL-3 + VisitShare (25%)	uct-100	0.688	[0.458, 0.861]	+0.201	8.9
BASIL-3 + UnreachStop (25%)	uct-100	0.701	[0.542, 0.854]	+0.215	9.0
BASIL-3 + Pre-VOC (25%)	uct-100	0.632	[0.403, 0.847]	+0.146	9.9
BASIL-3 + Disagreement (25%)	uct-100	0.528	[0.333, 0.708]	+0.042	7.7
BASIL-3 + Confidence (25%)	uct-100	0.507	[0.277, 0.736]	+0.021	11.5
BASIL-3 + RandomGate (25%)	uct-100	0.549	[0.375, 0.722]	+0.062	6.9
OracleSim (Full Search)	uct-100	0.708	[0.542, 0.847]	+0.222	36.0
OracleSim + LTH-Regret (10%)	uct-100	0.514	[0.333, 0.681]	+0.028	3.2
OracleSim + LTH-Regret (25%)	uct-100	0.653	[0.444, 0.833]	+0.167	9.7
OracleSim + LTH-Regret (50%)	uct-100	0.708	[0.535, 0.868]	+0.222	18.1

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